PRIME DIVISORS AND COMPONENTS OF EFFECTIVE CARTIER DIVISORS HAVE DELTA-DIMENSION ONE LESS THAN THE AMBIENT SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: an unpacking of the definitions for the first part and, for the second, an etale neighbourhood cutting the divisor by a nonzerodivisor together with the principal ideal theorem

Editorial note. The label lemma-divisor-delta-dimension names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The definition of a prime divisor on an integral good algebraic space and of the associated Weil divisor, and the identification of codimension one points with the local rings of dimension one
- The definition of a dimension function and its compatibility with codimension, so that codimension one means delta-dimension one less
- The local description of an effective Cartier divisor as the zero scheme of a nonzerodivisor on an etale neighbourhood, and Krull's principal ideal theorem

Source reference. Original lemma 00KV: sources/stacks-chow/tags.txt, source lines 765–765. Prerequisite pointer invoked at proof line(s) 1627; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0ED1: sources/stacks-chow/tags.txt, source lines 17976–17976. Prerequisite pointer invoked at proof line(s) 1613; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0ABV: sources/stacks-chow/tags.txt, source lines 13033–13033. Prerequisite pointer invoked at proof line(s) 1625; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 1594; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 083C: sources/stacks-chow/tags.txt, source lines 10282–10282. Prerequisite pointer invoked at proof line(s) 1623; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 0ENJ: sources/stacks-chow/tags.txt, source lines 18344–18344. Prerequisite pointer invoked at proof line(s) 1612; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 02I9: sources/stacks-chow/tags.txt, source lines 3292–3292. Prerequisite pointer invoked at proof line(s) 1615; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let X/B be good. Assume X is integral.*

(1) *If $Z \subset X$ is an integral closed subspace, then the following are equivalent:*
   (a) *Z is a prime divisor,*
   (b) *$|Z|$ has codimension 1 in $|X|$, and*
   (c) $\dim_\delta(Z) = \dim_\delta(X) - 1$.

(2) *If Z is an irreducible component of an effective Cartier divisor on X, then* $\dim_\delta(Z) = \dim_\delta(X) - 1$.

Proof. Part (1) follows from the definition of a prime divisor (Spaces over Fields, Definition 0ENJ), Decent Spaces, Lemma 0ED1, and the definition of a dimension function (Topology, Definition 02I9).

Let $D \subset X$ be an effective Cartier divisor. Let $Z \subset D$ be an irreducible component and let $\xi \in |Z|$ be the generic point. Choose an étale neighbourhood $(U, u) \to (X, \xi)$ where $U = \mathrm{Spec}(A)$ and $D \times_X U$ is cut out by a nonzerodivisor $f \in A$, see Divisors on Spaces, Lemma 083C. Then u is a generic point of $V(f)$ by Decent Spaces, Lemma 0ABV. Hence $\mathcal{O}_{U,u}$ has dimension 1 by Krull's Hauptidealsatz (Algebra, Lemma 00KV). Thus ξ is a codimension 1 point on X and Z is a prime divisor as desired. □